

14 Bound States in External Fields

In our calculations of radiative corrections in Chapter 11 we went just one step beyond the lowest order in perturbation theory. However, there is a very important class of problems where even the simplest calculation requires that from the beginning we consider classes of Feynman diagrams of arbitrarily high order in coupling constants like e . These problems are those involving bound states—in electrodynamics, either ordinary atoms and molecules, or such exotic atoms as positronium or muonium.

It is easy to see that such problems necessarily involve a breakdown of ordinary perturbation theory. Consider for instance the amplitude for electron–proton scattering as a function of the center-of-mass energy E . As shown in Section 10.3, the existence of a bound state like the ground state of hydrogen implies the existence of a pole in this amplitude at $E = m_p + m_e - 13.6$ eV. However, no single term in the perturbation series for electron–proton scattering has such a pole. The pole therefore can only arise from a divergence of the sum over all diagrams at center-of-mass energies near $m_p + m_e$.

The reason for this divergence of the perturbation series is also easy to see, especially if for the moment we consider the time-ordered diagrams of old-fashioned perturbation theory instead of Feynman diagrams. Suppose that in the center-of-mass system the electron and proton both have momenta of magnitude $q \ll m_e$, and consider an intermediate state in which the electron and proton momenta are different but also of order q . The energy denominator factor contributed by this state will be of order $[q^2/m_e]^{-1}$. Each such state will also contribute a matrix element of the Coulomb interaction of order e^2/q^2 (the Fourier transform of e^2/r), and the corresponding momentum space integration will contribute a factor of order q^3 . Putting this together, we see that each additional Coulomb interaction contributes an overall factor of order

$$[q^2/m_e]^{-1}[e^2/q^2][q^3] = e^2 m_e/q.$$

Thus the perturbation theory should break down when q is less than or of the order of $e^2 m_e$, or in other words when the kinetic and potential energies, which are of order q^2/m_e , are no larger than about $e^4 m_e$, which, of course, is of the order of the binding energy of hydrogen.

Our problem here is to learn how to use perturbation theory to evaluate radiative corrections in bound-state problems, summing to all orders those diagrams that need to be summed to all orders, and keeping only a finite number of those that don't.

14.1 The Dirac Equation

We shall limit ourselves in this chapter to problems in which bound states arise because of the Coulomb interaction of electrons (or muons) with heavy charged particles such as atomic nuclei. As shown in Section 13.6, this interaction may be taken into account by adding to the interaction Lagrangian a term¹ representing the effects of a c-number external vector

¹In this chapter we return to the use of an upper case Ψ to denote the electron field in the Heisenberg picture, reserving a lower case ψ for the Dirac field with time-dependence governed solely by the c-number external field $\mathcal{A}^\mu(x)$.

potential $\mathcal{A}^\mu(x)$:

$$\begin{aligned}\mathcal{L}_{\mathcal{A}} = & -ie\bar{\Psi}\gamma^\mu\Psi\mathcal{A}_\mu - \frac{1}{4}(Z_3 - 1)(\partial^\mu\mathcal{A}^\nu - \partial^\nu\mathcal{A}^\mu)(\partial_\mu A_\nu - \partial_\nu A_\mu) \\ & - ie(Z_2 - 1)\mathcal{A}_\mu\bar{\Psi}\gamma^\mu\Psi,\end{aligned}\tag{14.1.1}$$

which is obtained by replacing the quantum vector potential A^μ with $A^\mu + \mathcal{A}^\mu$ in the interaction part of Eq. (11.1.6). For instance, for a single heavy particle of charge Ze at the origin,

$$\mathcal{A}^0(x) = \frac{Ze}{4\pi|\mathbf{x}|}, \quad \mathbf{A}(x) = 0.\tag{14.1.2}$$

It is the interaction (14.1.1) that must be taken into account to all orders. In this section we will consider the theory with only this interaction, leaving radiative corrections to subsequent sections.

Physicists learn in kindergarten to approach this sort of problem by solving the wave equation of Dirac in the presence of the external field. It might seem unnecessary to derive this equation here, but as emphasized in Chapter 1, Dirac's original motivation for this equation as a sort of relativistic Schrödinger equation does not stand up to inspection. Also, in the course of our derivation we will discover the normalization conditions that have to be imposed on the solutions of the Dirac equation, which seemed somewhat *ad hoc* in Dirac's approach. The solutions of the Dirac equation discussed here will be important ingredients in our treatment of radiative corrections in the next section.

We will work here in a version of the Heisenberg picture, in which the time-dependence of operators is determined by a Hamiltonian including the external field interaction (14.1.1), but no other interactions. The electron field $\psi(x)$ in this picture satisfies the field equation

$$\left[\gamma^\lambda \frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda \mathcal{A}_\lambda(x)\right] \psi(x) = 0.\tag{14.1.3}$$

This is not the Dirac equation in the original sense of Dirac [1], because $\psi(x)$ here is not a c-number wave function but a quantum operator. The c-number Dirac wave functions are defined by

$$u_N(x) \equiv \langle \Omega | \psi(x) | N \rangle,\tag{14.1.4}$$

$$v_N(x) \equiv \langle N | \psi(x) | \Omega \rangle,\tag{14.1.5}$$

where the $|N\rangle$ are a complete orthonormal set of state-vectors, with $|\Omega\rangle$ the vacuum. It follows immediately from Eq. (14.1.3) that these functions satisfy the homogeneous Dirac equation

$$\begin{aligned}\left[\gamma^\lambda \frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda \mathcal{A}_\lambda(x)\right] u_N(x) &= \left[\gamma^\lambda \frac{\partial}{\partial x^\lambda} + m + ie\gamma^\lambda \mathcal{A}_\lambda(x)\right] v_N(x) \\ &= 0.\end{aligned}\tag{14.1.6}$$

We can also derive a normalization condition from the equal-time anticommutation relations for the Dirac field. These are unaffected by the interaction (14.1.1), and therefore take the same form as for the free fields:

$$\{\psi(\mathbf{x}, t), \bar{\psi}(\mathbf{y}, t)\} = i\gamma^0 \delta^3(\mathbf{x} - \mathbf{y}).\tag{14.1.7}$$

Taking the vacuum expectation value and inserting a sum over the states $|N\rangle$, we find

$$\sum_N u_N(\mathbf{x}, t) u_N^\dagger(\mathbf{y}, t) + \sum_N v_N(\mathbf{x}, t) v_N^\dagger(\mathbf{y}, t) = \delta^3(\mathbf{x} - \mathbf{y}), \quad (14.1.8)$$

it being understood that the sum over N includes an integral over continuum states as well as a sum over any discrete bound states.

We are chiefly interested in the case of a time-independent external field, like (14.1.2). In such cases, the states $|N\rangle$ may be taken as eigenstates of the Hamiltonian (including the interaction (14.1.1)) with energies E_N . Time-translation invariance then tells us that the $u_N(x)$ and $v_N(x)$ have the time-dependence:

$$u_N(\mathbf{x}, t) = e^{-iE_N t} u_N(\mathbf{x}), \quad v_N(\mathbf{x}, t) = e^{+iE_N t} v_N(\mathbf{x}). \quad (14.1.9)$$

The homogeneous Dirac equations (14.1.6) then become

$$i\gamma^0 [\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m + ie\gamma^\lambda \mathcal{A}_\lambda(\mathbf{x})] u_N(\mathbf{x}) = E_N u_N(\mathbf{x}), \quad (14.1.10)$$

$$i\gamma^0 [\boldsymbol{\gamma} \cdot \boldsymbol{\nabla} + m + ie\gamma^\lambda \mathcal{A}_\lambda(\mathbf{x})] v_N(\mathbf{x}) = -E_N v_N(\mathbf{x}). \quad (14.1.11)$$

The minus sign on the right-hand side of Eq. (14.1.11) shows that the v_N are the famous ‘negative-energy’ solutions of Dirac. As shown by Eq. (14.1.8), these negative-energy solutions are needed to make up a complete set of wave functions. Of course, for moderate external fields there are no negative-energy states in the theory, so all E_N are positive, but there is still an important difference between the states with non-vanishing u_N or v_N : the definitions (14.1.4) and (14.1.5) show that a state can have $u_N \neq 0$ or $v_N \neq 0$ only if it has charge $-e$ or $+e$, respectively. It is in this sense that negative-energy solutions of the Dirac equation have something to do with the existence of antiparticles. However, this argument has nothing to do with the details of the Dirac equation, or even with the spin of the electron.

From the Dirac wave equations (14.1.10) and (14.1.11) we can easily see that wave functions of different energy are orthogonal. That is,

$$(E_M - E_N^*)(u_N^\dagger u_M) = \boldsymbol{\nabla} \cdot (u_N^\dagger i\gamma^0 \boldsymbol{\gamma} u_M),$$

so if $|\mathbf{x}|^2(u_N^\dagger i\gamma^0 \boldsymbol{\gamma} u_M)$ remains bounded as $|\mathbf{x}| \rightarrow 0$ and $|\mathbf{x}| \rightarrow \infty$, then

$$\int d^3x (u_N^\dagger(\mathbf{x}) u_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq E_M^*. \quad (14.1.12)$$

With similar boundary conditions for the v_N , we find in the same way that

$$\int d^3x (v_N^\dagger(\mathbf{x}) v_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq E_M^*, \quad (14.1.13)$$

$$\int d^3x (u_N^\dagger(\mathbf{x}) v_M(\mathbf{x})) = 0 \quad \text{if} \quad E_N \neq -E_M^*. \quad (14.1.14)$$

Taking $N = M$, Eqs. (14.1.12) and (14.1.13) tell us that the energies are all real. Dropping the complex conjugation of E_M in Eqs. (14.1.12)–(14.1.14), we see that u s of different

energy are orthogonal, *vs* of different energy are orthogonal, and (as long as the potential is not strong enough to produce negative energy states) all *us* are orthogonal to all *vs*. By a suitable choice of the discrete quantum numbers that characterize the states along with the energy, we can then always arrange that

$$\int d^3x (u_N^\dagger(\mathbf{x})u_M(\mathbf{x})) = 0 \quad \text{if } N \neq M, \quad (14.1.15)$$

$$\int d^3x (v_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0 \quad \text{if } N \neq M, \quad (14.1.16)$$

$$\int d^3x (u_N^\dagger(\mathbf{x})v_M(\mathbf{x})) = 0. \quad (14.1.17)$$

Multiplying Eq. (14.1.8) on the right with $u_M(\mathbf{y})$ or $v_M(\mathbf{y})$, we find then that these wave functions must satisfy the normalization conditions

$$\int d^3y (u_N^\dagger(\mathbf{y})u_M(\mathbf{y})) = \int d^3y (v_N^\dagger(\mathbf{y})v_M(\mathbf{y})) = \delta_{NM}, \quad (14.1.18)$$

where δ_{NM} is a product of Kronecker deltas and momentum space delta functions, with normalization adapted to that used in defining $\sum_N$, in such a way that $\sum_N \delta_{NM} = 1$. These normalization conditions have nothing directly to do with any probabilistic interpretation of the Dirac wave functions, but arise instead from the anticommutation relations (14.1.7) for the fields.

Let us now specialize to the case of a pure electrostatic external field with $\mathbf{A} = 0$. In our standard representation of the Dirac matrices, we have

$$\gamma = i \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad i\gamma^0 = \beta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

where $\boldsymbol{\sigma}$ is the usual three-vector of $2 \cdot 2$ Pauli matrices, and ‘ $\mathbf{1}$ ’ and ‘ 0 ’ here are the $2 \cdot 2$ unit and zero matrices. We introduce two-component wave functions f_N and g_N by setting

$$u_N = \frac{1}{\sqrt{2}} \begin{pmatrix} f_N + ig_N \\ f_N - ig_N \end{pmatrix}. \quad (14.1.19)$$

The energy eigenvalue condition (14.1.10) then takes the form:

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})f_N = (E_N + e\mathcal{A}^0 + m)g_N, \quad (14.1.20)$$

$$(\boldsymbol{\sigma} \cdot \boldsymbol{\nabla})g_N = -(E_N + e\mathcal{A}^0 - m)f_N. \quad (14.1.21)$$

In the non-relativistic case where $e\mathcal{A}^0 r \approx Z\alpha \ll 1$ the binding energy $m - E_N$ is of order $Z^2\alpha^2 m$, while the gradient operator is of the order of $Z\alpha m$, so g_N is smaller than f_N by a factor of order $Z\alpha$. (To find the positron wave functions v_N we replace E_N everywhere by $-E_N$, so in this case f_N is smaller than g_N by the same factor.) We shall return to this non-relativistic case at the end of this section.

Physical states may be classified as even or odd under space inversion:

$$P|N\rangle = \eta_N|N\rangle, \quad (14.1.22)$$

where η_N is a sign factor, ± 1 . Recall that with the intrinsic parity of the electron defined to be $+1$, the Dirac field has the space-inversion property

$$P\psi(\mathbf{x}, t)P^{-1} = \beta\psi(-\mathbf{x}, t),$$

so Eqs. (14.1.4) and (14.1.22) show that the Dirac wave functions satisfy the parity condition

$$u_N(\mathbf{x}) = \eta_N \beta u_N(-\mathbf{x}), \quad (14.1.23)$$

or equivalently

$$f_N(\mathbf{x}) = \eta_N f_N(-\mathbf{x}), \quad g_N(\mathbf{x}) = -\eta_N g_N(-\mathbf{x}). \quad (14.1.24)$$

Note that the parity of the state is the same as the parity of $f_N(\mathbf{x})$, not $g_N(\mathbf{x})$.

Where the potential $\mathcal{A}^0$ is rotationally invariant, the solutions of the wave equations here may be classified according to their total angular momentum j and parity η . For a given j , the components f and g may be expanded in spherical harmonics with orbital angular momentum $\ell = j + \frac{1}{2}$ and $\ell = j - \frac{1}{2}$, but for a definite parity $\eta = (-1)^{j \mp \frac{1}{2}}$, Eqs. (14.1.24) show that we can only have $\ell = j \mp \frac{1}{2}$ in f and $\ell = j \pm \frac{1}{2}$ in g . The usual rules of angular-momentum addition then show that for a state of total angular momentum j , total angular-momentum z -component μ , and parity $(-1)^{j \mp \frac{1}{2}}$, the ‘large’ two-component wave function f has the form

$$f(\mathbf{x}) = \begin{pmatrix} C_{j \mp \frac{1}{2}, \frac{1}{2}}(j \mu; \mu - \frac{1}{2}, \frac{1}{2}) Y_{j \mp \frac{1}{2}}^{\mu - \frac{1}{2}}(\hat{\mathbf{x}}) \\ C_{j \mp \frac{1}{2}, \frac{1}{2}}(j \mu; \mu + \frac{1}{2}, -\frac{1}{2}) Y_{j \mp \frac{1}{2}}^{\mu + \frac{1}{2}}(\hat{\mathbf{x}}) \end{pmatrix} F(|\mathbf{x}|), \quad (14.1.25)$$

where C and Y are the usual Clebsch–Gordan coefficients and spherical harmonics [2]. Also, given any wave function of definite total angular momentum and parity we can construct another wave function with the same j and μ but opposite parity by applying the operator $\boldsymbol{\sigma} \cdot \hat{\mathbf{x}}$, so the ‘small’ components may be put in the form

$$g(\mathbf{x}) = (\boldsymbol{\sigma} \cdot \hat{\mathbf{x}}) \begin{pmatrix} C_{j \pm \frac{1}{2}, \frac{1}{2}}(j \mu; \mu - \frac{1}{2}, \frac{1}{2}) Y_{j \pm \frac{1}{2}}^{\mu - \frac{1}{2}}(\hat{\mathbf{x}}) \\ C_{j \pm \frac{1}{2}, \frac{1}{2}}(j \mu; \mu + \frac{1}{2}, -\frac{1}{2}) Y_{j \pm \frac{1}{2}}^{\mu + \frac{1}{2}}(\hat{\mathbf{x}}) \end{pmatrix} G(|\mathbf{x}|). \quad (14.1.26)$$

It is conventional to define the orbital angular momentum ℓ of the state as the orbital angular momentum of the ‘large’ components $f(\mathbf{x})$,

$$\ell = j \mp \frac{1}{2}, \quad (14.1.27)$$

so that the parity is always $(-1)^\ell$.

Inserting Eqs. (14.1.25) and (14.1.26) in Eqs. (14.1.20) and (14.1.21) yields the coupled differential equations

$$\frac{dG}{dr} + \frac{k+1}{r}G + (E + e\mathcal{A}^0 - m)F = 0, \quad (14.1.28)$$

$$\frac{dF}{dr} - \frac{k-1}{r}F - (E + e\mathcal{A}^0 + m)G = 0, \quad (14.1.29)$$

where for parity $\eta = (-1)^{j \mp \frac{1}{2}}$,

$$k = \pm \left(j + \frac{1}{2} \right). \quad (14.1.30)$$

Let us now concentrate on the simple Coulomb field (14.1.2), for which $e\mathcal{A}^0 = Z\alpha/r$. The treatment of the Dirac equation in this case [3] is familiar, so it will be summarized briefly here just for completeness. It is easy to see that the solutions near the origin go as r^{s-1} , with $s^2 = k^2 - Z^2\alpha^2$. (Note that $k^2 \geq 1$, so the exponent s is real for $Z\alpha \leq 1$.) We must reject the solutions with $s < 0$ as being inconsistent with the normalization condition (14.1.18). The condition that the wave functions do not blow up for $r \rightarrow \infty$ then fixes the allowed values of the energy eigenvalues:

$$E_{n,j} = m \left[1 + \frac{Z^2\alpha^2}{\left[n - j - \frac{1}{2} + \sqrt{(j + \frac{1}{2})^2 - Z^2\alpha^2} \right]^2} \right]^{-1/2}, \quad (14.1.31)$$

where n is a ‘principal quantum number’ with

$$j + \frac{1}{2} \leq n. \quad (14.1.32)$$

It is noteworthy that these energies do not depend on the parity or ℓ , but only on n and j . For each n and j there are two solutions, corresponding to the two signs of k or the two possible parities, except that for $n = j + \frac{1}{2}$ we only have $k > 0$ and parity $(-1)^{j-\frac{1}{2}}$, so that $\ell = j - \frac{1}{2}$. With Eq. (14.1.32), this is the same as the familiar non-relativistic restriction that $\ell \leq n - 1$.

For light atoms with $Z\alpha \ll 1$, Eq. (14.1.31) yields the power series

$$E = m \left[1 - \frac{Z^2\alpha^2}{2n^2} + \frac{Z^4\alpha^4}{n^4} \left(\frac{3}{8} - \frac{n}{2j+1} \right) + \cdots \right]. \quad (14.1.33)$$

The first two terms, of course, just represent the rest energy and the binding energy as given by the non-relativistic Schrödinger equation. The leading term that depends on j as well as n is the third term, the first relativistic correction. For $n = 1$ there is only one value of the total angular momentum, $j = \frac{1}{2}$, and since here $n = j + \frac{1}{2}$ there is also only one parity, $(-1)^{j-\frac{1}{2}} = +1$, corresponding to $\ell = 0$. It is therefore difficult to see the effects of the relativistic corrections in Eq. (14.1.33) in the $n = 1$ states of hydrogenic atoms, though as we shall see in Section 14.3, this has recently become possible. On the other hand, for $n = 2$ we have a $j = \frac{1}{2}$ state with both parities (i.e., $2s_{1/2}$ and $2p_{1/2}$) as well as a $2p_{3/2}$ state with $j = \frac{3}{2}$ and negative parity. Eq. (14.1.33) gives the splitting between the p states in hydrogen as

$$E(2p_{3/2}) - E(2p_{1/2}) = \frac{\alpha^4 m_e}{32} = 4.5283 \cdot 10^{-5} \text{ eV}. \quad (14.1.34)$$

Such relativistic line splitting is known as the *fine structure* of the atomic state. From the beginning it was known that this prediction is in good agreement with the observed fine structure. On the other hand, the Dirac equation does not yield any energy difference in the

$2s_{1/2}$ and $2p_{1/2}$ states, so this is a good place to look for the effects of further corrections, to be considered in Section 14.3.

Before closing this section we shall consider the approximate forms for the wave functions and matrix elements in the non-relativistic case for a general electrostatic potential $\mathcal{A}^0$. (For a Coulomb potential, this is the limit $Z\alpha \ll 1$.) Since here $E_N + m \simeq 2m \gg |e\mathcal{A}^0|$, the ‘small’ components of the electron wave function are given approximately in terms of the large components by

$$g_N \simeq (\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}) f_N / 2m_e. \quad (14.1.35)$$

Eq. (14.1.21) then becomes just the non-relativistic Schrödinger equation

$$\left[-\frac{\nabla^2}{2m_e} - e\mathcal{A}^0 \right] f_N \simeq (E_N - m) f_N. \quad (14.1.36)$$

Since there is no longer any coupling between spin and orbital degrees of freedom in the equation for f_N , we may find a complete set of solutions of this equation in the form

$$f_N = \chi_N \psi_N(\mathbf{x}),$$

where χ_N is a two-component constant spinor, and $\psi_N(\mathbf{x})$ is an ordinary one-component solution of the Schrödinger equation. However, we often work with states that have definite values of the total angular momentum j , for which f_N is (for non-vanishing orbital angular momentum) a sum of such terms.

In the non-relativistic approximation, the four-component Dirac wave function takes the form

$$u_N \simeq \frac{1}{\sqrt{2}} \begin{bmatrix} (1 + i\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}/2m) f_N \\ (1 - i\boldsymbol{\sigma} \cdot \boldsymbol{\nabla}/2m) f_N \end{bmatrix} \quad (14.1.37)$$

and Eq. (14.1.18) gives the normalization condition

$$\int d^3x (f_N^\dagger f_M) \simeq \delta_{NM} - \frac{1}{4} (v^2)_{NM}, \quad (14.1.38)$$

where

$$(v^2)_{NM} \equiv -\frac{1}{m_e^2} \int d^3x f_N^\dagger(\mathbf{x}) \nabla^2 f_M(\mathbf{x}).$$

In relating matrix elements in an external field to free-particle matrix elements, it is useful to note that the momentum space wave function in an energy eigenstate N may be written

$$u_N(\mathbf{p}) \equiv (2\pi)^{-3/2} \int d^3p e^{-i\mathbf{p} \cdot \mathbf{x}} u_N(\mathbf{x}) \simeq \sum_{\sigma} u(\mathbf{p}, \sigma) [f_N(\mathbf{p})]_{\sigma}, \quad (14.1.39)$$

where $u(\mathbf{p}, \sigma)$ is the free-particle Dirac spinor

$$u(\mathbf{p}, \sigma) \simeq \frac{1}{\sqrt{2}} \begin{bmatrix} (1 - \mathbf{p} \cdot \boldsymbol{\sigma}/2m_e) \chi_{\sigma} \\ (1 + \mathbf{p} \cdot \boldsymbol{\sigma}/2m_e) \chi_{\sigma} \end{bmatrix}, \quad \chi_{+\frac{1}{2}} \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_{-\frac{1}{2}} \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and $f_N(\mathbf{p})$ is the Fourier transform of the two-component Schrödinger wave function

$$f_N(\mathbf{p}) \equiv (2\pi)^{-3/2} \int d^3p e^{-i\mathbf{p} \cdot \mathbf{x}} f_N(\mathbf{x}).$$

In closing, as an aid in calculating the effects of various perturbations, we note that the leading terms in the electron matrix elements of the sixteen independent 4×4 matrices are

$$(\bar{u}_M u_N) \simeq (f_M^\dagger f_N) - \frac{1}{4m_e^2} (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \boldsymbol{\sigma} \cdot \nabla f_N), \quad (14.1.40)$$

$$i(\bar{u}_M \gamma^0 u_N) \simeq (f_M^\dagger f_N) + \frac{1}{4m_e^2} (\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \boldsymbol{\sigma} \cdot \nabla f_N), \quad (14.1.41)$$

$$(\bar{u}_M \gamma^i u_N) \simeq \frac{1}{2m_e} \left[(\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \sigma_i f_N) - (f_M^\dagger \sigma_i \boldsymbol{\sigma} \cdot \nabla f_N) \right], \quad (14.1.42)$$

$$(\bar{u}_M [\gamma^0, \gamma^i] u_N) \simeq \frac{i}{m_e} \left[(\nabla f_M^\dagger \cdot \boldsymbol{\sigma} \sigma_i f_N) + (f_M^\dagger \sigma_i \boldsymbol{\sigma} \cdot \nabla f_N) \right], \quad (14.1.43)$$

$$(\bar{u}_M [\gamma^i, \gamma^j] u_N) \simeq 2i\epsilon_{ijk} (f_M^\dagger \sigma_k f_N), \quad (14.1.44)$$

$$(\bar{u}_M \gamma^5 \gamma^i u_N) \simeq -i(f_M^\dagger \sigma_i f_N), \quad (14.1.45)$$

$$(\bar{u}_M \gamma^5 \gamma^0 u_N) \simeq \frac{1}{2m_e} \left[(\nabla f_M^\dagger \cdot \boldsymbol{\sigma} f_N) - (f_M^\dagger \boldsymbol{\sigma} \cdot \nabla f_N) \right], \quad (14.1.46)$$

$$(\bar{u}_M \gamma^5 u_N) \simeq \frac{i}{2m_e} \left[(\nabla f_M^\dagger \cdot \boldsymbol{\sigma} f_N) + (f_M^\dagger \boldsymbol{\sigma} \cdot \nabla f_N) \right]. \quad (14.1.47)$$

14.2 Radiative Corrections in External Fields

We now consider radiative corrections to the results of the previous section, due to the interaction of electrons with the quantum electromagnetic field as well as the external field of the heavy charged particles. These radiative corrections can be calculated using Feynman diagrams of the usual sort, with the whole effect of the external field being to modify the propagator of the electron field in the presence of the external electromagnetic field (and to supply the external-field-dependent renormalization counterterms shown in Eq. (14.1.1)). To be specific, the effect of inserting any number of vertices corresponding to the first term of the interaction (14.1.1) in an internal electron line of any graph is to replace the bare coordinate space propagator $-iS(x-y)$ with a corrected propagator

$$\begin{aligned} -iS_{\mathcal{A}}(x, y) \equiv & -iS(x-y) + (-i)^2 \int d^4 z_1 S(x-z_1) e\gamma^\mu \mathcal{A}_\mu(z_1) S(z_1-y) \\ & + (-i)^3 \int d^4 z_1 \int d^4 z_2 S(x-z_1) e\gamma^\mu \mathcal{A}_\mu(z_1) S(z_1-z_2) e\gamma^\nu \mathcal{A}_\nu(z_2) S(z_2-y) + \cdots, \end{aligned} \quad (14.2.1)$$

where as usual

$$S(x-y) \equiv \frac{1}{(2\pi)^4} \int d^4 p \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} e^{ip \cdot (x-y)}.$$

(We must write $S_{\mathcal{A}}$ as a function of x and y rather than of $x-y$, because the external field invalidates translation invariance.) The theorem proved in Section 6.4 tells us that Eq. (14.2.1) is the same as

$$-iS_{\mathcal{A}}(x, y) = \mathcal{A} \langle \Omega | T \{ \psi(x), \bar{\psi}(y) \} | \Omega \rangle_{\mathcal{A}}. \quad (14.2.2)$$

with the subscript $\mathcal{A}$ on the right indicating that the vacuum state $|\Omega\rangle_{\mathcal{A}}$ and electron field $\psi(x)$ are to be defined in a Heisenberg picture in which the only interaction taken into account is the interaction (14.1.1) with the external field. Inserting a complete set of intermediate states $|N\rangle_{\mathcal{A}}$ in Eq. (14.2.2) yields an expression for the propagator in terms of the Dirac wave functions u_N and v_N introduced in the previous section

$$-iS_{\mathcal{A}}(x, y) = \theta(x^0 - y^0) \sum_N u_N(x) \bar{u}_N(y) - \theta(y^0 - x^0) \sum_M v_M(x) \bar{v}_M(y). \quad (14.2.3)$$

It is also possible to obtain the propagator (14.2.2) as the solution of the *inhomogeneous Dirac equation*:

$$\left[\gamma^\mu \frac{\partial}{\partial x^\mu} + m + ie\gamma^\mu \mathcal{A}_\mu(x) \right] S_{\mathcal{A}}(x, y) = \delta^4(x - y), \quad (14.2.4)$$

which follows from the field equations (14.1.3) and anticommutation relations (14.1.7), or formally from the perturbation series (14.2.1). Also Eq. (14.2.3) tells us that the propagator satisfies boundary conditions; its Fourier decomposition contains only ‘positive frequency terms’ proportional to $\exp\{-iE(x^0 - y^0)\}$ with $E > 0$ for $x^0 - y^0 \rightarrow \infty$, and only ‘negative frequency terms’ proportional to $\exp\{+iE(x^0 - y^0)\}$ with $E > 0$ for $x^0 - y^0 \rightarrow -\infty$. The inhomogeneous Dirac equation with these boundary conditions may be used to obtain a numerical solution [4] for this propagator even in cases where the external field is too strong to allow the use of the perturbation series (14.2.1). Once the propagator $S_{\mathcal{A}}(x, y)$ has been calculated, the amplitudes for scattering in an external field can be calculated using ordinary Feynman diagrams, but with $S_{\mathcal{A}}(x, y)$ in place of $S(x - y)$ (and with $\mathcal{A}$ -dependent renormalization counterterms inserted where appropriate).

Now let us see how to use the perturbation series with this corrected propagator to calculate the shifts of bound-state energy levels. Consider the full electron propagator $S'_{\mathcal{A}}(x, y)$, involving interactions of the electron with the quantum electromagnetic field as well as the external field:

$$-iS'_{\mathcal{A}}(x, y) = {}_{\mathcal{A}}\langle \Omega | T \{ \Psi(x), \bar{\Psi}(y) \} | \Omega \rangle_{\mathcal{A}} \quad (14.2.5)$$

with $\Psi(x)$ the electron field in a Heisenberg picture including all interactions, and $|\Omega\rangle$ the vacuum eigenstate of the full Hamiltonian. For a time-independent external potential we can find a complete orthonormal set $|N\rangle$ of eigenstates of the full Hamiltonian with energies E'_N . Inserting a sum over these states in the operator product in Eq. (14.2.5), we find

$$\begin{aligned} -iS'_{\mathcal{A}}(x, y) &= \theta(x^0 - y^0) e^{-iE'_N(x^0 - y^0)} \sum_N U_N(\mathbf{x}) \bar{U}_N(\mathbf{y}) \\ &\quad - \theta(y^0 - x^0) e^{-iE'_N(y^0 - x^0)} \sum_N V_N(\mathbf{x}) \bar{V}_N(\mathbf{y}), \end{aligned} \quad (14.2.6)$$

where

$$\langle \Omega | \Psi(\mathbf{x}, t) | N \rangle \equiv e^{-iE'_N t} U_N(\mathbf{x}), \quad (14.2.7)$$

$$\langle N | \bar{\Psi}(\mathbf{x}, t) | \Omega \rangle \equiv e^{+iE'_N t} \bar{V}_N(\mathbf{x}). \quad (14.2.8)$$

(The sum includes an integral over continuum states as well as a sum over discrete bound states. As before, U_N and V_N are non-zero only if the state $|N\rangle$ has charge $-e$ or $+e$, respectively.) We can redefine the propagator as a function of energy rather than time

$$S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) \equiv \int_{-\infty}^{\infty} dx^0 e^{iE(x^0 - y^0)} S'_{\mathcal{A}}(x, y). \quad (14.2.9)$$

(Time-translation invariance dictates that $S'_{\mathcal{A}}(x, y)$ is a function of $x^0 - y^0$ but not of x^0 and y^0 separately.) From Eq. (14.2.6) we see that

$$S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \sum_N \frac{U_N(\mathbf{x}) \bar{U}_N(\mathbf{y})}{E'_N - E - i\epsilon} - \sum_N \frac{V_N(\mathbf{x}) \bar{V}_N(\mathbf{y})}{E'_N + E - i\epsilon}. \quad (14.2.10)$$

In particular, $S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E)$ has a pole at any electron bound-state energy, and also at the negative of any positron bound-state energy. (Of course, positrons do not have bound states in the Coulomb field of an ordinary positively charged nucleus.)

Let us now consider the lowest-order radiative corrections to the complete propagator. The Feynman rules here give the complete propagator to this order as $S'_{\mathcal{A}} = S_{\mathcal{A}} + \delta S_{\mathcal{A}}$, with a correction term

$$\delta S_{\mathcal{A}}(x, y) = \int d^4z \int d^4w S_{\mathcal{A}}(x, z) \Sigma_{\mathcal{A}}^*(z, w) S_{\mathcal{A}}(w, y), \quad (14.2.11)$$

where $i\Sigma_{\mathcal{A}}^*$ is the sum of all one-loop diagrams with one incoming and one outgoing electron line (excluding final electron propagators) calculated using $S_{\mathcal{A}}(x, y)$ in place of $S(x - y)$ for internal electron lines, plus second-order renormalization counterterms. Using energy variables in place of time variables, this is

$$\delta S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \int d^3z \int d^3w S_{\mathcal{A}}(\mathbf{x}, \mathbf{z}; E) \Sigma_{\mathcal{A}}^*(\mathbf{z}, \mathbf{w}; E) S_{\mathcal{A}}(\mathbf{w}, \mathbf{y}; E), \quad (14.2.12)$$

where

$$\Sigma_{\mathcal{A}}^*(\mathbf{z}, \mathbf{w}; E) \equiv \int dz^0 e^{iE(z^0 - w^0)} \Sigma_{\mathcal{A}}^*(z, w). \quad (14.2.13)$$

The effect of these radiative corrections is to change the wave functions to $U_N = u_N + \delta u_N$ and $V_N = v_N + \delta v_N$ and the bound-state energies to $E'_N = E_N + \delta E_N$, so that the complete propagator is

$$\begin{aligned} S'_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) &\simeq S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) \\ &+ \sum_N \frac{\delta u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) + u_N(\mathbf{x}) \delta \bar{u}_N(\mathbf{y})}{E_N - E} \\ &- \sum_N \frac{\delta v_N(\mathbf{x}) \bar{v}_N(\mathbf{y}) + v_N(\mathbf{x}) \delta \bar{v}_N(\mathbf{y})}{E_N + E} \\ &- \sum_N \frac{u_N(\mathbf{x}) \bar{u}_N(\mathbf{y}) \delta E_N}{(E_N - E)^2} + \sum_N \frac{v_N(\mathbf{x}) \bar{v}_N(\mathbf{y}) \delta E_N}{(E_N + E)^2}. \end{aligned} \quad (14.2.14)$$

(We are dropping the $i\epsilon$ terms because we are now not taking E to lie in the continuum of scattering states.) We see that the shift δE_N of an electron bound-state energy is given by

the coefficient of $-u_N(\mathbf{x})\bar{u}_N(\mathbf{y})/(E_N - E)^2$ in the complete propagator. To calculate this, we note that Eq. (14.2.3) gives

$$S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) = \sum_N \frac{u_N(\mathbf{x})\bar{u}_N(\mathbf{y})}{E_N - E - i\epsilon} - \sum_N \frac{v_N(\mathbf{x})\bar{v}_N(\mathbf{y})}{E_N + E - i\epsilon}. \quad (14.2.15)$$

Inserting this in Eq. (14.2.12) gives

$$\begin{aligned} \delta S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E) &= \sum_{N,M} \frac{u_N(\mathbf{x})\bar{u}_M(\mathbf{y})}{(E_N - E)(E_M - E)} \\ &\quad \cdot \int d^3z \int d^3w \bar{u}_N(\mathbf{z})\Sigma_{\mathcal{A}}^*(\mathbf{z}, \mathbf{w}; E)u_M(\mathbf{w}) + \dots \end{aligned} \quad (14.2.16)$$

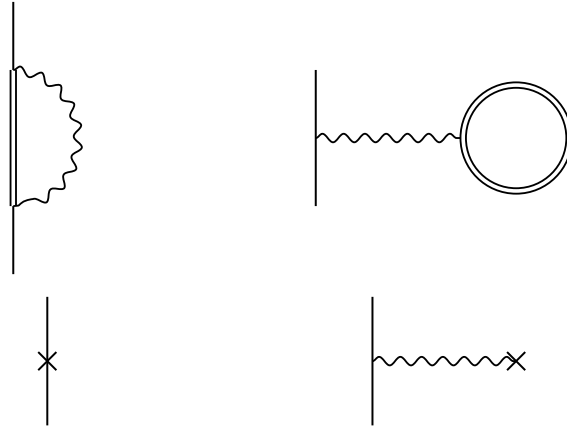


Figure 14.1. Lowest-order Feynman diagrams for the electron self-energy function $\Sigma_{\mathcal{A}}^*(x, y)$ in the presence of an external field. Here double straight lines represent electron propagators $S_{\mathcal{A}}$ that include effects of the external field; single straight lines are the incoming and outgoing electron lines; wavy lines are virtual photons; the cross represents renormalization counterterms.

where the dots denote additional terms involving at least one negative-energy pole. Comparing the coefficients of $(E_N - E)^{-2}$ here and in Eq. (14.2.14), we find

$$\delta E_N = - \int d^3x \int d^3y \bar{u}_N(\mathbf{x})\Sigma_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E_N)u_N(\mathbf{y}). \quad (14.2.17)$$

The u_N are solutions of the homogeneous Dirac equation satisfying the normalization condition (14.1.8), so this is very much like ordinary first-order perturbation theory, but with $-\Sigma^*$ in place of a perturbation to the Hamiltonian.

Generally δE_N turns out to be complex. This is simply a consequence of the instability of atomic energy levels to radiative decay to lower levels; we saw in Chapter 3 that an unstable state of energy E and decay rate Γ produces poles in various amplitudes at the complex energy $E - i\Gamma/2$. The imaginary part of Eq. (14.2.17) therefore equals $-\Gamma/2$, while its real part gives the energy shift.

The Feynman diagrams for Σ^* are shown in Figure 14.1. (Note that there are new photon tadpole diagrams here because the external field breaks the Lorentz invariance and

charge-conjugation invariance that forbids such diagrams in the ordinary Feynman rules.) Application of the position space Feynman rules to these diagrams gives

$$\begin{aligned}
i\Sigma_{\mathcal{A}}^*(x, y) = & [e\gamma^\mu][-iS_{\mathcal{A}}(x, y)][e\gamma_\mu][-iD(x - y)] \\
& - [e\gamma^\mu\delta^4(x - y)] \int d^4z [-iD(x - z)] \text{Tr}\{[-iS_{\mathcal{A}}(z, z)][e\gamma_\mu]\} \\
& - i(Z_2 - 1)(\gamma^\mu\partial_\mu + m)\delta^4(x - y) + i\delta m\delta^4(x - y) \\
& + e\gamma^\mu(Z_2 - 1)\delta^4(x - y)\mathcal{A}_\mu(x) \\
& + i(Z_3 - 1)[e\gamma_\mu]\delta^4(x - y) \int d^4z [-iD(x - z)]\partial_\nu(\partial^\nu\mathcal{A}^\mu(z) - \partial^\mu\mathcal{A}^\nu(z)).
\end{aligned} \tag{14.2.18}$$

with the renormalization constants $(Z_2 - 1)$, $(Z_3 - 1)$, and δm calculated to second order in e . (The minus sign in the second term is the usual one that accompanies closed fermion loops.)

For strong external fields with $Z\alpha$ of order unity it is necessary to calculate the configuration space electron propagator $S_{\mathcal{A}}$ and the integrals in Eqs. (14.2.17) and (14.2.18) numerically [4]. However, for weak fields we can use the first few terms of the series (14.2.1) in Eq. (14.2.18) and calculate these integrals in closed form. For this purpose, it is more convenient to work in momentum space, defining:

$$S_{\mathcal{A}}(x, y) = (2\pi)^{-4} \int d^4p' d^4p e^{ip'\cdot x} e^{-ip\cdot y} S_{\mathcal{A}}(p', p), \tag{14.2.19}$$

$$\Sigma_{\mathcal{A}}^*(x, y) = (2\pi)^{-4} \int d^4p' d^4p e^{ip'\cdot x} e^{-ip\cdot y} \Sigma_{\mathcal{A}}^*(p', p), \tag{14.2.20}$$

$$u_N(\mathbf{x}) = (2\pi)^{-3/2} \int d^3p e^{i\mathbf{p}\cdot\mathbf{x}} u_N(\mathbf{p}), \tag{14.2.21}$$

$$\mathcal{A}^\mu(x) = \int d^4q e^{iq\cdot x} \mathcal{A}^\mu(q). \tag{14.2.22}$$

(We are here committing the impropriety of using the same symbol for a function and its Fourier transform, leaving it to the displayed arguments of the functions to indicate which is which.) Then Eqs. (14.2.1) and (14.2.18) become

$$S_{\mathcal{A}}(p', p) = \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} - ie \frac{-i\not{p}' + m}{p'^2 + m^2 - i\epsilon} \mathcal{A}(p' - p) \frac{-i\not{p} + m}{p^2 + m^2 - i\epsilon} + \dots \tag{14.2.23}$$

and

$$\begin{aligned}
\Sigma_{\mathcal{A}}^*(p', p) = & \frac{ie^2}{(2\pi)^4} \int \frac{d^4k}{k^2 - i\epsilon} \gamma^\mu S_{\mathcal{A}}(p' - k, p - k) \gamma_\mu \\
& + [-(Z_2 - 1)(i\not{p} + m) + Z_2\delta m] \delta^4(p' - p) - ie(Z_2 - 1) \mathcal{A}(p' - p) \\
& - \frac{ie^2\gamma^\mu}{(2\pi)^4[(p - p')^2 - i\epsilon]} \int d^4q \text{Tr}\{S_{\mathcal{A}}(q, q + p' - p) \gamma_\mu\} \\
& + \frac{ie(Z_3 - 1)}{(p - p')^2 - i\epsilon} [(p - p')^2 \mathcal{A}(p' - p) - (\not{p} - \not{p}')(p - p') \cdot \mathcal{A}(p' - p)].
\end{aligned} \tag{14.2.24}$$

Because the external field is time-independent, $S_{\mathcal{A}}(x, y)$ and $\Sigma_{\mathcal{A}}^*(x, y)$ can depend on x^0 and y^0 only through the difference $x^0 - y^0$, so $S_{\mathcal{A}}(p', p)$ and $\Sigma_{\mathcal{A}}^*(p', p)$ as well as $\mathcal{A}^\mu(p' - p)$ must be proportional to $\delta(p'^0 - p^0)$:

$$\mathcal{A}^\mu(p' - p) = \delta(p'^0 - p^0) \mathcal{A}^\mu(\mathbf{p}' - \mathbf{p}), \quad (14.2.25)$$

$$S_{\mathcal{A}}(p', p) = \delta(p'^0 - p^0) S_{\mathcal{A}}(\mathbf{p}', \mathbf{p}; p^0), \quad (14.2.26)$$

$$\Sigma_{\mathcal{A}}^*(p', p) = \delta(p'^0 - p^0) \Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; p^0). \quad (14.2.27)$$

The energy shift is then given by Eqs. (14.2.17) and (14.2.13) as

$$\delta E_N = - \int d^3 p' \int d^3 p \bar{u}_N(\mathbf{p}') \Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; E_N) u_N(\mathbf{p}), \quad (14.2.28)$$

with $\Sigma_{\mathcal{A}}^*(\mathbf{p}', \mathbf{p}; E_N)$ given by Eqs. (14.2.23), (14.2.24), and (14.2.27). This is the master formula we shall use in the next section to calculate energy shifts in weak external fields.

14.3 The Lamb Shift in Light Atoms

Let us now consider radiative corrections to the energy levels of a non-relativistic electron in a general electrostatic field, such as an electron in the Coulomb field of a light nucleus with $Z\alpha \ll 1$. It is natural in this limit to treat the Coulomb field as a weak perturbation, but we will see that this would lead to an infrared divergence, related to that found in Section 11.3. The infrared divergence is really fictitious, because the four-momenta $\mathbf{p}, E_N$ and $\mathbf{p}', E_N$ are not on the electron mass shell, but it does force us to proceed with some care.

Usually this problem is dealt with by dividing the integral over virtual photon energies into a low-energy range, within which we can treat the electrons non-relativistically but must include effects to all orders in the external field, and a high-energy range, within which we have to include relativistic effects but may include only effects of lowest order in the external field. Instead, we shall here introduce a fictitious photon mass μ , chosen to be much larger than typical electron kinetic energies, but much less than typical electron momenta. For a Coulomb field, this amounts to the requirement that

$$(Z\alpha)^2 m_e \ll \mu \ll Z\alpha m_e. \quad (14.3.1)$$

We write the photon propagator in the first two terms of Eq. (14.2.24) (including the formulas for the counterterms $Z_2 - 1$ and $Z_2 \delta m$) in the form

$$\frac{1}{k^2 - i\epsilon} = \left[\frac{1}{k^2 + \mu^2 - i\epsilon} \right] + \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right]. \quad (14.3.2)$$

The energy shift is correspondingly a sum of two terms, a ‘high-energy’ and a ‘low-energy’ term: The high-energy term is calculated by using the first term in the photon propagator (14.3.2) in the first three terms in Eq. (14.2.24), and adding the result to the last two terms (the vacuum polarization terms) in Eq. (14.2.24) which are not infrared divergent anyway; the low-energy term is calculated using the second term in Eq. (14.3.2) in the first three

terms of Eq. (14.2.24). One advantage of this procedure is that we shall be able to use the results of the relativistic calculations of Sections 11.3 and 11.4 directly, without the rather tricky conversion from a photon mass to an infrared energy cutoff. Of course, in the end we shall have to check that the dependence on the photon mass μ in the high-energy and low-energy contributions to the energy shift cancel, leaving the total energy shift μ -independent.

A High-Energy Term

Because μ is taken much larger than the atomic binding energies we can here keep only terms of lowest order in the external field. The one-loop radiative correction to atomic energy levels in a general time-independent external vector potential $\mathcal{A}^\mu(\mathbf{x})$ is given in momentum space by Eq. (14.2.28), with the self-energy insertion $\Sigma_{\mathcal{A}}(p', p)$ given by Eqs. (14.2.24) and (14.2.23). The terms of zeroth order in the external field simply cancel: the δm term cancels the first term with $\mathcal{A} = 0$; the $Z_3 - 1$ term cancels the third term with $\mathcal{A} = 0$; and the $Z_2 - 1$ terms vanish because $u(p)$ satisfies the Dirac equation. The term in $\Sigma_{\mathcal{A}}(p', p)$ that is of first order in $\mathcal{A}^\mu$ may be put in the form

$$\Sigma_{\mathcal{A}1}(p', p) = -ie\mathcal{A}_\mu(p' - p)\Gamma_1^\mu(p', p) \quad (14.3.3)$$

with

$$\begin{aligned} \Gamma_1^\mu(p', p) = & \frac{ie^2}{(2\pi)^4} \int \frac{d^4k}{k^2 + \mu^2 - i\epsilon} \left[\frac{-i(\not{p}' - \not{k}) + m_e}{(p' - k)^2 + m_e^2 - i\epsilon} \right] \gamma^\mu \left[\frac{-i(\not{p} - \not{k}) + m_e}{(p - k)^2 + m_e^2 - i\epsilon} \right] + (Z_2 - 1)\gamma^\mu \\ & - \frac{ie^2\gamma_\nu}{(2\pi)^4[(p - p')^2 - i\epsilon]} \int d^4l \operatorname{Tr} \left\{ \left[\frac{-i\not{l} + m_e}{l^2 + m_e^2} \right] \gamma^\mu \left[\frac{-i\not{l} - i\not{p}' + i\not{p} + m_e}{(l + p' - p)^2 + m_e^2} \right] \gamma^\nu \right\} \\ & - \frac{Z_3 - 1}{(p' - p)^2 - i\epsilon} [(p - p')^2 \eta_{\mu\nu} - (p' - p)_\mu(p' - p)_\nu] \gamma^\nu. \end{aligned} \quad (14.3.4)$$

Comparison of the first two terms with Eqs. (11.3.1) and (11.3.8) and the second two terms with Eqs. (11.3.9), (11.2.3), and (11.2.15) reveals that $\Gamma_1^\mu(p', p)$ is the complete one-loop vertex function, including vacuum polarization and all counterterms, whose mass-shell matrix elements we have already calculated in Section 11.3. Using Eqs. (14.2.26) and (14.2.25), this contribution to the energy shift (14.2.28) is given by

$$[\delta E_N]_{\text{high energy}} = ie \int d^3p' \int d^3p (\bar{u}_N(\mathbf{p}')\Gamma_1^\mu(\mathbf{p}', E_N, \mathbf{p}, E_N)u_N(\mathbf{p}))\mathcal{A}_\mu(\mathbf{p}' - \mathbf{p}). \quad (14.3.5)$$

(This could have been guessed at, by simply replacing the γ^μ in the interaction of the electron with the external field with Γ_1^μ .) As discussed in Section 14.1, because $Z\alpha \ll 1$ we can approximate the Dirac wave function u_N in Eq. (14.3.5) as

$$[u_N(\mathbf{p})]_\alpha = \sum_\sigma u_\alpha(\mathbf{p}, \sigma)[f_N(\mathbf{p})]_\sigma, \quad (14.3.6)$$

where f_N is the non-relativistic two-component wave function of an electron in the external Coulomb field, and $u(\mathbf{p}, \sigma)$ is the four-component normalized solution of the momentum space Dirac equation

$$[i\gamma_\mu p^\mu + m_e]u(\mathbf{p}, \sigma) = 0 \quad (14.3.7)$$

for spin z -component σ . Since $u_N(\mathbf{p})$ approximately satisfies the free-particle Dirac equation, Eq. (10.6.15) gives the general form of the matrix element of Γ_1^μ as

$$\bar{u}_N(\mathbf{p}')[\gamma^\mu + \Gamma_1^\mu(p', p)]u_N(\mathbf{p}) = \bar{u}_N(\mathbf{p}') \left[\gamma^\mu F_1(q^2) + \frac{i}{2}[\gamma^\mu, \gamma^\nu]q_\nu F_2(q^2) \right] u_N(\mathbf{p}), \quad (14.3.8)$$

where $q \equiv p' - p$. The wave functions $u_N(\mathbf{p})$ fall off very rapidly for $|\mathbf{p}| \gg Z\alpha m_e$, so we only need $F_1(q^2)$ and $F_2(q^2)$ in the limit $|q^2| \ll m_e^2$. In this limit, Eqs. (11.3.31), (10.6.18), and (11.3.16) give

$$F_1(q^2) \simeq 1 + \frac{e^2}{24\pi^2} \left(\frac{q^2}{m_e^2} \right) \left[\ln \left(\frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right], \quad (14.3.9)$$

$$F_2(q^2) \simeq \frac{e^2}{16m_e\pi^2}. \quad (14.3.10)$$

Let us first consider the contribution of the F_1 term in Eq. (14.3.8), which makes by far the largest contribution to the energy shift, and raises the most interesting problems in its calculation. For a pure electrostatic field with $\mathcal{A} = 0$, Eqs. (14.3.5), (14.3.8), and (14.3.9) give

$$[\delta E_N]_{F_1} = -\frac{e^2}{24\pi^2 m_e^2} \left[\ln \left(\frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \int d^3 p' \int d^3 p \bar{u}_N(\mathbf{p}') (-ie\mathcal{A}^0(\mathbf{p}' - \mathbf{p})) \gamma^0(\mathbf{p}' - \mathbf{p})^2 u_N(\mathbf{p}). \quad (14.3.11)$$

To calculate this contribution, we may use the leading term in the non-relativistic matrix element (14.1.41), and find

$$[\delta E_N]_{F_1} = -\frac{e^2}{24\pi^2 m_e^2} \left[\ln \left(\frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \int d^3 p' \int d^3 p f_N^\dagger(\mathbf{p}') e\mathcal{A}^0(\mathbf{p}' - \mathbf{p}) [\mathbf{p}' - \mathbf{p}]^2 f_N(\mathbf{p}) \quad (14.3.12)$$

or in position space

$$[\delta E_N]_{F_1} = \frac{e^2}{24\pi^2 m_e^2} \left[\ln \left(\frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \int d^3 x f_N^\dagger(\mathbf{x}) [e\nabla^2 \mathcal{A}^0(\mathbf{x})] f_N(\mathbf{x}). \quad (14.3.13)$$

In particular, for the Coulomb potential (14.1.2), we have $e\nabla^2 \mathcal{A}^0(\mathbf{x}) = -Ze^2\delta^3(\mathbf{x})$, and the label N consists of a principal quantum number n and angular-momentum quantum numbers j, m, ℓ , while (11.2.41) gives $[f_{njm\ell}(0)]_\sigma = 2(Z\alpha m_e/n)^{3/2} \delta_{\ell,0} \delta_{\sigma,m}/\sqrt{4\pi}$. The energy shift (14.3.11) is then

$$[\delta E_{nj\ell}]_{F_1} = -\frac{2Z^4\alpha^5 m_e}{3\pi n^3} \left[\ln \left(\frac{\mu^2}{m_e^2} \right) + \frac{2}{5} + \frac{3}{4} \right] \delta_{\ell,0}. \quad (14.3.14)$$

(The lack of dependence of δE on the total angular momentum z -component m is guaranteed by rotational invariance.) The term $2/5$ in the brackets in Eqs. (14.3.12) and (14.3.13) arises from vacuum polarization, and yields just the energy shift calculated somewhat heuristically in Section 11.2.

Before going on to calculate the magnetic and low-energy contributions to the energy shift, it is worth noting that the result we have obtained so far yields a fair order-of-magnitude estimate of the Lamb shift without further work. We can anticipate that the low-energy terms will contain a term proportional to $\ln(\mu/B)$, with a coefficient such as to cancel the μ -dependence in Eq. (14.3.12). The constant B here is an energy which must be included to make the argument of the logarithm dimensionless; since it is the binding of the electron in the atom that will eventually provide our infrared cutoff, we may guess that B is a typical atomic binding energy, of order $B \simeq (Z\alpha)^2 m_e$. The total energy shift in a state N with principal quantum number n and orbital angular momentum ℓ is therefore of the form

$$\delta E_N = -\frac{2Z^4 \alpha^5 m_e}{3\pi n^3} [\ln(Z^4 \alpha^4) \delta_{\ell,0} + O(1)]. \quad (14.3.15)$$

For the $2s$ state of hydrogen, the logarithmic term alone gives

$$\delta E_{2s} = -\frac{\alpha^5 m_e}{12\pi} \ln(\alpha^4) = 5.5 \cdot 10^{-6} \text{ eV} = 1300 \text{ MHz} \cdot 2\pi\hbar.$$

As we shall see, the ‘ $O(1)$ ’ terms in Eq. (14.3.15) will lower the total energy shift by about 25%.

Now let us consider the contribution of the F_2 term in the matrix element of Γ_1^μ , which as we saw in Section 10.6 may be interpreted as a radiative correction to the magnetic moment of the electron. Using Eqs. (14.3.10), (14.3.8), and (14.3.6) in Eq. (14.3.5), we find that this term gives an energy shift

$$[\delta E_N]_{F_2} = -\frac{e^2}{32\pi^2 m_e} \int d^3 p' \int d^3 p (\bar{u}_N(\mathbf{p}') [\gamma^\mu, \gamma^\nu] u_N(\mathbf{p})) e \mathcal{A}_\mu(\mathbf{p}' - \mathbf{p}) (p' - p)_\nu. \quad (14.3.16)$$

or in position space

$$[\delta E_N]_{F_2} = \frac{ie^2}{64\pi^2 m_e} \int d^3 x (\bar{u}_N(\mathbf{x}) [\gamma^\mu, \gamma^\nu] u_N(\mathbf{x})) e \mathcal{F}_{\mu\nu}(\mathbf{x}), \quad (14.3.17)$$

where

$$\mathcal{F}_{\mu\nu}(\mathbf{x}) = \partial_\mu \mathcal{A}_\nu(\mathbf{x}) - \partial_\nu \mathcal{A}_\mu(\mathbf{x}). \quad (14.3.18)$$

For a pure electrostatic field with $\mathcal{A} = 0$, this is

$$[\delta E_N]_{F_2} = -\frac{ie^2}{32\pi^2 m_e} \int d^3 x (\bar{u}_N(\mathbf{x}) [\gamma^0, \gamma] u_N(\mathbf{x})) \cdot \nabla [e \mathcal{A}^0(\mathbf{x})]. \quad (14.3.19)$$

In the non-relativistic limit $Z\alpha \ll 1$, we can use the approximate result (14.1.43), which here reads

$$\begin{aligned} (\bar{u}_N [\gamma^0, \gamma] u_N) &\simeq \frac{i}{m_e} \left[(\nabla f_N^\dagger) \cdot \boldsymbol{\sigma} \boldsymbol{\sigma} f_N + f_N^\dagger \boldsymbol{\sigma} \boldsymbol{\sigma} \cdot \nabla f_N \right] \\ &= \frac{i}{m_e} \left[\nabla (f_N^\dagger f_N) - i(\nabla f_N^\dagger \times \boldsymbol{\sigma}) f_N - i f_N^\dagger (\boldsymbol{\sigma} \times \nabla f_N) \right]. \end{aligned} \quad (14.3.20)$$

Using this in Eq. (14.3.8) and integrating by parts, this part of the energy shift is

$$[\delta E_N]_{F_2} = -\frac{e^2}{32\pi^2 m_e^2} \int d^3x \left[-|f_N(\mathbf{x})|^2 \nabla^2 (e\mathcal{A}^0(\mathbf{x})) + 2if_N^\dagger(\mathbf{x}) \boldsymbol{\sigma} \cdot (\nabla(e\mathcal{A}^0(\mathbf{x})) \times \nabla f_N(\mathbf{x})) \right]. \quad (14.3.21)$$

Combining Eqs. (14.3.12) and (14.3.21) gives the total high-energy contribution to the energy shift in an arbitrary electrostatic potential $\mathcal{A}^0$:

$$[\delta E_N]_{\text{high energy}} = \frac{e^2}{24\pi^2 m_e^2} \left[\ln \left(\frac{\mu^2}{m_e^2} \right) + \frac{2}{5} \right] \int d^3x f_N^\dagger(\mathbf{x}) [e\nabla^2 \mathcal{A}^0(\mathbf{x})] f_N(\mathbf{x}) + \frac{ie^2}{16\pi^2 m_e^2} \int d^3x f_N^\dagger(\mathbf{x}) \boldsymbol{\sigma} \cdot (\nabla[e\mathcal{A}^0(\mathbf{x})] \times \nabla f_N(\mathbf{x})). \quad (14.3.22)$$

B Low-Energy Term

The low-energy contribution to the energy shift is obtained from the first three terms in Eq. (14.2.24), making the replacement in the photon propagator

$$\frac{1}{k^2 - i\epsilon} \longrightarrow \frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon}. \quad (14.3.23)$$

This substitution will eventually serve to cut off the integral over components of the photon four-momentum k at values of order μ , but it is not possible to see this until we carefully take mass renormalization into account, so we shall defer making any non-relativistic approximations until then. Also, we are now including photon momenta as small or smaller than the binding energies of the atomic states, so we must treat the electrostatic forces responsible for this binding to all orders.

Instead of working with the momentum space formula (14.2.24), it will be convenient to return to the configuration space formula (14.2.18). This gives the low-energy contribution to the electron self-energy function as

$$[\Sigma_{\mathcal{A}}^*(x, y)]_{\text{low energy}} = ie^2 \gamma^\mu S_{\mathcal{A}}(x, y) \gamma_\mu D(x - y; \mu) + \delta m_e(\mu) \delta^4(x - y) - i(Z_2(\mu) - 1) \{ \gamma^\mu [\partial_\mu + ie\mathcal{A}_\mu] + m_e \} \delta^4(x - y), \quad (14.3.24)$$

where $D(x - y; \mu)$ is the modified photon propagator

$$D(x - y; \mu) = \frac{1}{(2\pi)^4} \int d^4k e^{-ik \cdot (x-y)} \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right], \quad (14.3.25)$$

and the counterterms $Z_2(\mu) - 1$ and $\delta m(\mu)$ are calculated using this modified propagator. Converting from time to energy variables, the low-energy contribution to the function (14.2.13) is then

$$[\Sigma_{\mathcal{A}}^*(\mathbf{x}, \mathbf{y}; E)]_{\text{low energy}} = \frac{ie^2}{(2\pi)^4} \int d^4k \gamma^\mu S_{\mathcal{A}}(\mathbf{x}, \mathbf{y}; E - k^0) \gamma_\mu \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right] e^{i\mathbf{k} \cdot (\mathbf{x}-\mathbf{y})} - (Z_2(\mu) - 1) (\boldsymbol{\gamma} \cdot \nabla + i\gamma^0 E + ie\gamma^\nu \mathcal{A}_\nu + m_e) \delta^3(\mathbf{x} - \mathbf{y}) + \delta m_e(\mu) \delta^3(\mathbf{x} - \mathbf{y}). \quad (14.3.26)$$

The low-energy contribution to the energy shift is then given by Eq. (14.2.17) as

$$\begin{aligned}
[\delta E_N]_{\text{low energy}} &= - \int d^3x \int d^3y \bar{u}_N(\mathbf{x}) [\Sigma_A^*(\mathbf{x}, \mathbf{y}; E_N)]_{\text{low energy}} u_N(\mathbf{y}) \\
&= - \frac{ie^2}{(2\pi)^4} \int d^4k \int d^3x \int d^3y \bar{u}_N(\mathbf{x}) \gamma^\mu S_A(\mathbf{x}, \mathbf{y}; E_N - k^0) \gamma_\mu u_N(\mathbf{y}) \\
&\quad \cdot \left[\frac{1}{k^2 - i\epsilon} - \frac{1}{k^2 + \mu^2 - i\epsilon} \right] e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} - \delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}).
\end{aligned} \tag{14.3.27}$$

Note that the terms proportional to $Z_2(\mu) - 1$ have dropped out because the Dirac wave function $u_N(\mathbf{x})$ satisfies the Dirac equation Eq. (14.1.10). For the electron propagator in the presence of the Coulomb field, we use Eq. (14.2.15):

$$S_A(\mathbf{x}, \mathbf{y}; E) = \sum_M \frac{u_M(\mathbf{x}) \bar{u}_M(\mathbf{y})}{E_M - E - i\epsilon} - \sum_M \frac{v_M(\mathbf{x}) \bar{v}_M(\mathbf{y})}{E_M + E - i\epsilon},$$

with the sums in the first and second terms running over all one-electron and one-positron states, respectively. The k^0 integrals can be done most easily by closing the contour of integration with a large semi-circle in the lower half-plane in the first term and in the upper half-plane in the second term:

$$\int dk^0 \left(\frac{1}{k^2 + \mu^2 - i\epsilon} \right) \left(\frac{1}{E_M \mp E_N \pm k^0 - i\epsilon} \right) = \frac{i\pi}{\sqrt{\mathbf{k}^2 + \mu^2}} \left(\frac{1}{E_M \mp E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon} \right),$$

and likewise if μ is replaced with zero. The energy shift (14.3.27) now becomes

$$\begin{aligned}
[\delta E_N]_{\text{low energy}} &= - \frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[\Gamma_{MN}^\rho(\mathbf{k})^* \Gamma_{\rho MN}(\mathbf{k}) \left(\frac{1}{|\mathbf{k}|(E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \right. \\
&\quad \left. \left. - \frac{1}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right) \right. \\
&\quad \left. - \tilde{\Gamma}_{MN}^\rho(\mathbf{k})^* \tilde{\Gamma}_{\rho MN}(\mathbf{k}) \left(\frac{1}{|\mathbf{k}|(E_M + E_N + |\mathbf{k}| - i\epsilon)} \right. \right. \\
&\quad \left. \left. - \frac{1}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M + E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right) \right] \\
&\quad - \delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}),
\end{aligned} \tag{14.3.28}$$

where

$$\Gamma_{MN}^\rho(\mathbf{k}) \equiv \int d^3y e^{-i\mathbf{k} \cdot \mathbf{y}} \bar{u}_M(\mathbf{y}) \gamma^\rho u_N(\mathbf{y}), \tag{14.3.29}$$

$$\tilde{\Gamma}_{MN}^\rho(\mathbf{k}) \equiv \int d^3y e^{-i\mathbf{k} \cdot \mathbf{y}} \bar{v}_M(\mathbf{y}) \gamma^\rho u_N(\mathbf{y}). \tag{14.3.30}$$

(Of course, the ‘sum’ over M in Eq. (14.3.28) receives contributions only from electron states in the first term and positron states in the second term.) Eq. (14.3.28) could have been derived more directly from old-fashioned perturbation theory; the energy denominators

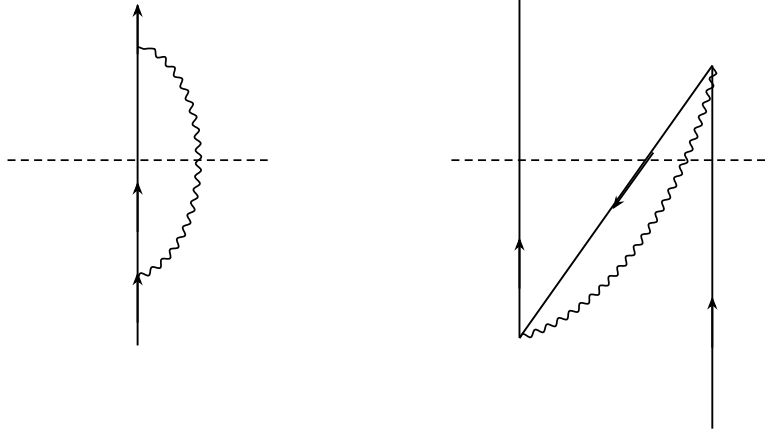


Figure 14.2. Old-fashioned perturbation theory diagrams for the low-energy part of the electron energy shift. Here solid lines are electrons; wavy lines are photons; the dashed line cuts through particle lines corresponding to the intermediate states appearing in the first two terms in (14.3.28)

$E_M - E_N + \omega$ and $E_M + E_N + \omega$ are the result of subtracting the energy E_N of the initial state from the energy of an intermediate state consisting of either an electron of energy E_M and a photon of energy ω , or else a positron of energy E_M , a photon of energy ω , and both the final and the initial electron. (See Figure 14.2.)

Before making any approximations to Eq. (14.3.28), it will be convenient to express the time components of the matrix elements Γ_{MN}^ρ and $\tilde{\Gamma}_{MN}^\rho$ in terms of the corresponding space components, using relations² derived from the conservation of the electric current:

$$k_i \Gamma_{MN}^i(\mathbf{k}) = (E_N - E_M) \Gamma_{MN}^0(\mathbf{k}), \quad (14.3.31)$$

$$k_i \tilde{\Gamma}_{MN}^i(\mathbf{k}) = (E_N + E_M) \tilde{\Gamma}_{MN}^0(\mathbf{k}). \quad (14.3.32)$$

Furthermore, by using the completeness relation (14.1.8) it is straightforward to show that

$$\sum_M \left[|\Gamma_{MN}^0(\mathbf{k})|^2 + |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 \right] = 1 \quad (14.3.33)$$

and

$$\begin{aligned} \sum_M \left[|\Gamma_{MN}^0(\mathbf{k})|^2 (E_M - E_N) - |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 (E_M + E_N) \right] &= \sum_M \left[-\Gamma_{MN}^{0*}(\mathbf{k}) \mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k}) \right. \\ &\quad \left. - \tilde{\Gamma}_{MN}^{0*}(\mathbf{k}) \mathbf{k} \cdot \tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k}) \right] \\ &= -i \mathbf{k} \cdot \int d^3x \bar{u}_N(\mathbf{x}) \boldsymbol{\gamma} u_N(\mathbf{x}) = 0, \end{aligned} \quad (14.3.34)$$

²To derive Eq. (14.3.31), note that

$$\begin{aligned} k_i \Gamma_{MN}^i(\mathbf{k}) &= -i \int d^3x e^{-i\mathbf{k} \cdot \mathbf{x}} \nabla \cdot (\bar{u}_M(\mathbf{x}) \boldsymbol{\gamma} u_N(\mathbf{x})) \\ &= i \int d^3x e^{-i\mathbf{k} \cdot \mathbf{x}} \partial_0 \left[\bar{u}_M(\mathbf{x}) \gamma^0 u_N(\mathbf{x}) e^{-i(E_N - E_M)x^0} \right]_{x^0=0} = (E_N - E_M) \Gamma_{MN}^0(\mathbf{k}). \end{aligned}$$

Eq. (14.3.32) is derived in the same way.

the last step following from the parity condition (14.1.23). In this way, Eq. (14.3.28) can be rewritten

$$\begin{aligned}
[\delta E_N]_{\text{low energy}} = & -\frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[\frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{|\mathbf{k}|(E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \\
& \left. - \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right] \\
& + \frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[\frac{|\tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{|\mathbf{k}|(E_M + E_N + |\mathbf{k}|)} \right. \\
& \left. - \frac{|\tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \tilde{\mathbf{\Gamma}}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{\sqrt{\mathbf{k}^2 + \mu^2}(E_M + E_N + \sqrt{\mathbf{k}^2 + \mu^2})} \right] \\
& - \frac{e^2}{(2\pi)^3} \int d^3k \sum_M |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 \left(\frac{1}{\mathbf{k}^2} - \frac{1}{\mathbf{k}^2 + \mu^2} \right) \\
& + \frac{1}{2} \alpha \mu - \delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}). \tag{14.3.35}
\end{aligned}$$

In the next-to-last term, we have used the elementary integral

$$\int d^3k \left(\frac{1}{\mathbf{k}^2} - \frac{1}{\mathbf{k}^2 + \mu^2} \right) = 2\mu\pi^2.$$

So far, this has been an exact rewriting of Eq. (14.3.28). We must now invoke several approximations. First, consider mass renormalization. We have already calculated $\delta m_e(\mu)$ to order α in Section 11.4; it is

$$\delta m_e(\mu) = \frac{2m_e\pi^2 e^2}{(2\pi)^4} \int_0^1 dx (1+x) \ln \left(\frac{m_e^2 x^2 + \mu^2(1-x)}{m_e^2 x^2} \right). \tag{14.3.36}$$

Although in Section 11.4 we regarded μ as a regulator mass, to be taken much larger than m_e , we can just as well use Eq. (14.3.36) to provide a value for $\delta m_e(\mu)$ in the case that interests us here, $\mu \ll m_e$. In this limit Eq. (14.3.36) gives

$$\delta m_e(\mu) \longrightarrow \frac{\alpha\mu}{2} \left[1 - \frac{3\mu}{2\pi m_e} + \dots \right]. \tag{14.3.37}$$

We also recall that for $Z\alpha \ll 1$, $u_N(\mathbf{x})$ is given by Eq. (14.1.37) as

$$u_N(\mathbf{x}) = \frac{1}{\sqrt{2}} \left[(1 - \boldsymbol{\sigma} \cdot \mathbf{v}/2 + \dots) f_N(\mathbf{x}) \right], \tag{14.3.38}$$

where dots indicate terms of higher order in $Z\alpha$; $\mathbf{v}$ is the non-relativistic velocity operator $-i\nabla/m_e$; and $f_N(\mathbf{x})$ is a two-component spinor solution of the non-relativistic Schrödinger equation, normalized according to Eq. (14.1.38) so that

$$\int |f_N(\mathbf{x})|^2 = 1 - \frac{1}{4}(\mathbf{v}^2)_{NN} + \dots. \tag{14.3.39}$$

This gives the coefficient of $\delta m_e(\mu)$ in Eq. (14.3.35) as

$$\int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x}) = 1 - \frac{1}{2}(\mathbf{v}^2)_{NN} + \dots \quad (14.3.40)$$

We note immediately that the leading term in $-\delta m_e(\mu) \int d^3x \bar{u}_N u_N$ cancels the term $\alpha\mu/2$ in Eq. (14.3.35). Indeed, we could have anticipated this cancellation, because the term $\alpha\mu/2$ in Eq. (14.3.35) survives in the limit $Z\alpha \rightarrow 0$, and the definition of $m_e(\mu)$ as the renormalized electron mass implies that there must be no energy shift in this limit. By the same argument we can anticipate that the term of order $\alpha\mu^2/m_e$ in $\delta m_e(\mu)$ (which is larger than of order $\alpha(Z\alpha)^4 m_e$, and therefore cannot simply be neglected) cancels the second and third terms in Eq. (14.3.35), which are also of this order.³ On the other hand, the product of the $\alpha\mu^2/m_e$ term in δm_e with the second term in the matrix element (14.3.40) is of order $(Z\alpha)^2 \alpha\mu^2/m_e \ll \alpha(Z\alpha)^4 m_e$, and may therefore be neglected. To order $\alpha(Z\alpha)^4 m_e$, the only remaining effect of mass renormalization is to leave us with the product of the leading term in $\delta m_e(\mu)$ with the term of order $(Z\alpha)^2$ in $\int d^3x \bar{u}_N u_N$:

$$-\left[\frac{e^2\mu}{8\pi}\right] \left[-\frac{1}{2}(\mathbf{v}^2)_{NN}\right] = \frac{e^2\mu}{16\pi}(\mathbf{v}^2)_{NN}.$$

(This is the effect of mass renormalization on the electron kinetic energy, mentioned in Section 1.3.) It turns out that this is just the negative of what the first term in Eq. (14.3.35) would be if we neglected the difference between energy levels. To see this, note that the

³The cancellation can be shown as follows. We anticipate that the second and third terms in Eq. (14.3.35) are small enough so that they can be evaluated using the extreme nonrelativistic approximation $\beta u_N(\mathbf{x}) = u_N(\mathbf{x})$ for the Dirac equation satisfied by $u_N(\mathbf{x})$. On the other hand, although the Coulomb force may be neglected in the positron wave functions $v_M(\mathbf{x})$, the sum over M in the third term receives important contributions from relativistic positrons, so we use the approximation $v_{\mathbf{p},\sigma}(\mathbf{x}) = (2\pi)^{-3/2} v(\mathbf{p},\sigma) e^{i\mathbf{p}\cdot\mathbf{x}}$, where $v(\mathbf{p},\sigma)$ is the positron spinor introduced in Section 5.5, normalized so that $\bar{v}(\mathbf{p},\sigma') v(\mathbf{p},\sigma) = \delta_{\sigma'\sigma}$. Thus the sums over M in the second and third terms of Eq. (14.3.35) are approximately given by

$$\begin{aligned} \frac{1}{2} \left[\sum_M \tilde{\Gamma}_{MN}^{i*}(\mathbf{k}) \tilde{\Gamma}_{MN}^j(\mathbf{k}) + (i \leftrightarrow j) \right] &\simeq \delta_{ij} \left(\frac{\sqrt{k^2 + m_e^2} + m_e}{2\sqrt{k^2 + m_e^2}} \right), \\ \sum_M |\tilde{\Gamma}_{MN}^0(\mathbf{k})|^2 &\simeq \left(\frac{\sqrt{k^2 + m_e^2} - m_e}{2\sqrt{k^2 + m_e^2}} \right). \end{aligned}$$

To leading order in μ/m_e , the second and third terms in Eq. (14.3.35) are then, respectively,

$$\begin{aligned} -\frac{e^2}{4m_e(2\pi)^3} \int d^3k \left[\frac{2}{k} - \frac{3 - k^2/(k^2 + \mu^2)}{\sqrt{k^2 + \mu^2}} \right] \left(\frac{\sqrt{k^2 + m_e^2} + m_e}{2\sqrt{k^2 + m_e^2}} \right) &\simeq \frac{\alpha\mu^2}{4\pi m_e}, \\ -\frac{e^2}{2(2\pi)^3} \int d^3k \left(\frac{1}{k^2} - \frac{1}{k^2 + \mu^2} \right) \left(\frac{\sqrt{k^2 + m_e^2} - m_e}{2\sqrt{k^2 + m_e^2}} \right) &\simeq -\frac{\alpha\mu^2}{\pi m_e}. \end{aligned}$$

(Eq. (14.3.32) rules out the possibility that a relativistic correction to the latter expression might not be suppressed by the factor k^2/m_e^2 that appears in this expression for $|\mathbf{k}|^2 \ll m_e^2$.) These two terms are cancelled by the term $+3\mu^2/4\pi m_e$ in $-\delta m_e(\mu) \int d^3x \bar{u}_N(\mathbf{x}) u_N(\mathbf{x})$. Finally, we note that relativistic corrections to the above estimates for the sums over positron states would involve additional factors of $v^2/c^2 \simeq (Z\alpha)^2$, yielding contributions of order $\alpha(Z\alpha)^2 \mu^2/m_e \ll \alpha(Z\alpha)^4 m_e$, which justifies the non-relativistic approximations used here.

integral in this term is effectively cut off at $|\mathbf{k}| \sim \mu \ll Z\alpha m_e$, so we can evaluate the matrix element $\mathbf{\Gamma}_{MN}(\mathbf{k})$ in the limit $\mathbf{k} \rightarrow 0$. To lowest order in $Z\alpha$, Eq. (14.1.42) gives

$$\mathbf{\Gamma}_{MN}(0) = (\mathbf{v})_{MN} \quad (14.3.41)$$

and using the completeness of the solutions f_N of the non-relativistic Schrödinger equation, we then have

$$\sum_M \Gamma_{MN}^{i*}(\mathbf{k}) \Gamma_{MN}^j(\mathbf{k}) \simeq (v^i v^j)_{NN}, \quad (14.3.42)$$

so to this order

$$\begin{aligned} & -\frac{e^2}{2(2\pi)^3} \int d^3k \sum_M \left[\frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{\mathbf{k}^2} - \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{\mathbf{k}^2 + \mu^2} \right] \\ & \simeq -\frac{e^2}{2(2\pi)^3} (\mathbf{v}^2)_{NN} \int d^3k \left[\frac{2}{3\mathbf{k}^2} - \frac{1 - \mathbf{k}^2/3(\mathbf{k}^2 + \mu^2)}{\mathbf{k}^2 + \mu^2} \right] = -\frac{e^2 \mu}{16\pi} (\mathbf{v}^2)_{NN}. \end{aligned}$$

Thus after mass renormalization we are left with just the first term in Eq. (14.3.35), less the same with energy differences $E_N - E_M$ dropped:

$$\begin{aligned} [\delta E_N]_{\text{low energy}} &= \frac{e^2}{2(2\pi)^3} \int d^3k \sum_M (E_M - E_N) \left[\frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / \mathbf{k}^2}{\mathbf{k}^2 (E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \\ & \quad \left. - \frac{|\mathbf{\Gamma}_{MN}(\mathbf{k})|^2 - |\mathbf{k} \cdot \mathbf{\Gamma}_{MN}(\mathbf{k})|^2 / (\mathbf{k}^2 + \mu^2)}{(\mathbf{k}^2 + \mu^2) (E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right]. \quad (14.3.43) \end{aligned}$$

Again using Eq. (14.3.41), this is

$$\begin{aligned} [\delta E_N]_{\text{low energy}} &= \frac{e^2}{2(2\pi)^3} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \int d^3k \left[\frac{2}{3\mathbf{k}^2 (E_M - E_N + |\mathbf{k}| - i\epsilon)} \right. \\ & \quad \left. - \frac{1 - \mathbf{k}^2/3(\mathbf{k}^2 + \mu^2)}{(\mathbf{k}^2 + \mu^2) (E_M - E_N + \sqrt{\mathbf{k}^2 + \mu^2} - i\epsilon)} \right]. \quad (14.3.44) \end{aligned}$$

Even though typical values of the electron momentum are much larger than typical atomic energy differences, this is not true of typical values of $|\mathbf{k}|$ in this integral, because the integral would be infrared divergent if we did not keep the $E_M - E_N$ terms in the denominators. The integral in Eq. (14.3.44) may be evaluated in the limit $\mu \gg |E_M - E_N| \sim (Z\alpha)^2 m_e$ by dividing the range of integration of $|\mathbf{k}|$ into two segments, from zero to λ and from λ to infinity, with λ chosen so that $|E_M - E_N| \ll \lambda \ll \mu$ but otherwise arbitrary. In this way we find

$$\begin{aligned} & \int_0^\infty k^2 dk \left[\frac{2}{3k^2 (E_M - E_N + k) - i\epsilon} - \frac{1 - k^2/3(k^2 + \mu^2)}{(k^2 + \mu^2) (E_M - E_N + \sqrt{k^2 + \mu^2} - i\epsilon)} \right] \\ & \simeq \frac{2}{3} \left[\ln \left(\frac{\mu}{2|E_M - E_N|} \right) + \frac{5}{6} + i\pi\theta(E_N - E_M) \right]. \end{aligned}$$

The imaginary term here reflects the possibility of decay of the atom in state N to states M of lower energy. This term contributes to the decay rate, given by the imaginary part

of the energy shift. We are interested here in the real part of the energy shift, and so will drop this imaginary term in what follows. Eq. (14.3.44) now gives

$$[\delta E_N]_{\text{low energy}} = \frac{e^2}{6\pi^2} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \left[\ln \left(\frac{\mu}{2|E_N - E_M|} \right) + \frac{5}{6} \right]. \quad (14.3.45)$$

C Total Energy Shift

We need to make a connection between the sum in Eq. (14.3.45) and the matrix element in the high-energy term (14.3.22). For this purpose, let's first see what value the sum in Eq. (14.3.45) would have if we could ignore the logarithm. We note that $(E_M - E_N)v_{NM} = [v, H]_{NM}$, so

$$\begin{aligned} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 &= \frac{1}{2} \sum_M ([v^i, H]_{NM} v_{MN}^i + v_{NM}^i [H, v^i]_{MN}) \\ &= -\frac{1}{2m_e^2} ([p^i, [p^i, H]])_{NN}. \end{aligned}$$

The only term in the non-relativistic Hamiltonian H that does not commute with the momentum operator $\mathbf{p}$ is the potential term $-e\mathcal{A}^0(\mathbf{x})$, so this gives

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 = -\frac{e}{2m_e^2} (\nabla^2 \mathcal{A}^0(\mathbf{x}))_{NN}. \quad (14.3.46)$$

Inspection of Eqs. (14.3.45) and (14.3.22) now shows that the term proportional to $\ln \mu$ in the high-energy term is cancelled by the term proportional to $\ln \mu$ in the low-energy term:

$$\begin{aligned} \delta E_N &= [\delta E_N]_{\text{high energy}} + [\delta E_N]_{\text{low energy}} \\ &= \frac{e^2}{6\pi^2} \sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \left[\ln \left(\frac{m_e}{2|E_N - E_M|} \right) + \frac{5}{6} - \frac{1}{5} \right] \\ &\quad - \frac{e^2}{16\pi^2 m_e^2} (\boldsymbol{\sigma} \cdot (\nabla(e\mathcal{A}^0(\mathbf{x})) \times \mathbf{p}))_{NN}. \end{aligned} \quad (14.3.47)$$

So far, this has been for a general electrostatic field $\mathcal{A}^0(\mathbf{x})$. Let us now specialize to a pure Coulomb field, with

$$\mathcal{A}^0(\mathbf{x}) = Ze/|\mathbf{x}|. \quad (14.3.48)$$

In this case, Eq. (14.3.46) reads

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 = \frac{Ze^2}{2m_e^2} (\delta^3(\mathbf{x}))_{NN} = \frac{Ze^2}{2m_e^2} (f_N^\dagger(0) f_N(0)). \quad (14.3.49)$$

This is non-vanishing only for $\ell = 0$. Also, the matrix element in the last term in Eq. (14.3.47) has the value

$$(\boldsymbol{\sigma} \cdot (\nabla(e\mathcal{A}^0(\mathbf{x})) \times \mathbf{p}))_{NN} = -Ze^2 \left(\frac{1}{r^3} \boldsymbol{\sigma} \cdot \mathbf{L} \right)_{NN}, \quad (14.3.50)$$

which is non-vanishing only for $\ell \neq 0$. It is therefore useful at this point to divide our consideration between the two cases, $\ell = 0$ and $\ell \neq 0$.

i. $\ell = 0$ It is convenient here to define a mean excitation energy ΔE_N :

$$\begin{aligned} \sum_M |\mathbf{v}_{MN}|^2 (E_M - E_N) \ln |E_N - E_M| &= \ln \Delta E_N \sum_M |\mathbf{v}_{MN}|^2 (E_M - E_N) \\ &= \frac{Ze^2}{2m_e^2} \ln \Delta E_N (f_N^\dagger(0) f_N(0)). \end{aligned} \quad (14.3.51)$$

For s -wave hydrogenic states, the label N consists of a principal quantum number n and spin z -component m , and $[f_{nm}(0)]_\sigma = 2(Z\alpha m_e/n)^{3/2} \delta_{\sigma,m}/\sqrt{4\pi}$, so

$$(f_N^\dagger(0) f_N(0)) = \frac{1}{\pi} \left(\frac{Z\alpha m_e}{n} \right)^3. \quad (14.3.52)$$

Using Eqs. (14.3.51) and (14.3.52) in Eq. (14.3.47) gives the energy shift in these states as

$$[\delta E]_{n,\ell=0} = \frac{4\alpha(Z\alpha)^4 m_e}{3\pi n^3} \left[\ln \left(\frac{m_e}{2\Delta E_{n,\ell=0}} \right) + \frac{19}{30} \right]. \quad (14.3.53)$$

ii. $\ell \neq 0$ For non-vanishing orbital angular momentum the sum (14.3.49) vanishes, so the definition (14.3.51) is inappropriate. Instead, it is conventional here to define a mean excitation energy ΔE_N by

$$\sum_M (E_M - E_N) |\mathbf{v}_{MN}|^2 \ln |E_N - E_M| = \frac{2(Z\alpha)^4 m_e}{n^3} \ln \left(\frac{2\Delta E_N}{Z^2 \alpha^2 m_e} \right). \quad (14.3.54)$$

(Because Eq. (14.3.49) vanishes, it makes no difference what units are used to measure $E_N - E_M$ in Eq. (14.3.54).) Also, in a state of total angular momentum j and orbital angular momentum ℓ , the scalar product $\boldsymbol{\sigma} \cdot \mathbf{L}$ has the familiar value $j(j+1) - \ell(\ell+1) - 3/4$, and for principal quantum number n the operator $1/r^3$ has the expectation value

$$\int d^3r |f_N|^2 / r^3 = \frac{2Z^3 \alpha^3 m_e^3}{n^3 \ell(\ell+1)(2\ell+1)}. \quad (14.3.55)$$

Putting this all together in Eq. (14.3.47), we have for $\ell \neq 0$:

$$\begin{aligned} [\delta E]_{nj\ell} &= - \frac{4\alpha(Z\alpha)^4 m_e}{3\pi n^3} \ln \left(\frac{2\Delta E_{nj\ell}}{Z^2 \alpha^2 m_e} \right) \\ &\quad + \frac{\alpha(Z\alpha)^4 m_e}{2\pi n^3} \left[\frac{j(j+1) - \ell(\ell+1) - \frac{3}{4}}{\ell(\ell+1)(2\ell+1)} \right]. \end{aligned} \quad (14.3.56)$$

It only remains to use these results to give numerical value for the energy shifts. The mean excitation energies here must be calculated numerically; using non-relativistic hydrogen wave functions, they have the values [5]:

$$\Delta E_{1s} = 19.769266917(6) \text{ Ry},$$

$$\Delta E_{2s} = 16.63934203(1) \text{ Ry},$$

$$\Delta E_{2p} = 0.9704293186(3) \text{ Ry},$$

where $1 \text{ Ry} \equiv m_e \alpha^2 / 2 = 13.6057 \text{ eV}$. Eq. (14.3.53) then gives

$$\begin{aligned} [\delta E]_{1s} &= \frac{4\alpha^5 m_e}{3\pi} \left[\ln \left(\frac{m_e}{2\Delta E_{1s}} \right) + \frac{19}{30} \right] = 3.3612 \cdot 10^{-5} \text{ eV} \\ &= 2\pi\hbar \cdot 8127.4 \text{ MHz}, \end{aligned} \quad (14.3.57)$$

$$\begin{aligned} [\delta E]_{2s} &= \frac{\alpha^5 m_e}{6\pi} \left[\ln \left(\frac{m_e}{2\Delta E_{2s}} \right) + \frac{19}{30} \right] = 4.2982 \cdot 10^{-5} \text{ eV} \\ &= 2\pi\hbar \cdot 1039.31 \text{ MHz}, \end{aligned} \quad (14.3.58)$$

while Eq. (14.3.56) gives

$$\begin{aligned} [\delta E]_{2p_{1/2}} &= \frac{\alpha^5 m_e}{6\pi} \left[\ln \left(\frac{\alpha^2 m_e}{2\Delta E_{2p}} \right) - \frac{1}{8} \right] = -5.3267 \cdot 10^{-8} \text{ eV} \\ &= 2\pi\hbar \cdot (-12.88) \text{ MHz}. \end{aligned} \quad (14.3.59)$$

The classic Lamb shift is the energy difference between the $2s$ and $2p_{1/2}$ states of the hydrogen atom, states that would be degenerate in the absence of radiative corrections. Our calculation has given

$$[\delta E]_{2s} - [\delta E]_{2p_{1/2}} = 4.35152 \cdot 10^{-5} \text{ eV} = 2\pi\hbar \cdot 1052.19 \text{ MHz}.$$

This is numerically close (though analytically not identical) to the old result of Kroll and Lamb [6] and French and Weisskopf [7], which they obtained using the techniques of old-fashioned perturbation theory. Earlier in this section we made a crude estimate of 1300 MHz by considering only the high-energy contribution to the $2s$ energy shift, with an infrared cutoff guessed to be of order $\alpha^2 m_e = 2 \text{ Ry}$. We can now see that this was an overestimate, arising mostly from the fact that the true value of the effective infrared cutoff $\Delta E_{2s} = 16.64 \text{ Ry}$ is considerably larger than we had guessed. On the other hand, as described in Section 1.3, in 1947 Hans Bethe [8] was able to make a rather good estimate of the Lamb shift, 1040 MHz, by considering only the low-energy contribution to the $2s$ energy shift, with an ultraviolet cutoff guessed to be m_e . (Bethe made the first estimate of the excitation energy, $\Delta E_{2s} \simeq 17.8 \text{ Ry}$.)

The calculation of the Lamb shift described here has been improved by the inclusion of higher-order radiative corrections and nuclear size and recoil effects. At present the greatest uncertainty is due to a doubt about the correct value of the rms charge radius r_p of the proton. For $r_p = 0.862 \cdot 10^{-13} \text{ cm}$ or $r_p = 0.805 \cdot 10^{-13} \text{ cm}$, one calculation [9] gives a Lamb shift of either 1057.87 MHz or 1057.85 MHz, while another [10] gives either 1057.883 MHz or 1057.865 MHz. Given the uncertainty in proton radius, the agreement is excellent with the present experimental value [11], 1057.845(9) MHz. The accuracy of this experimental value is limited chiefly by the $\sim 100 \text{ MHz}$ natural linewidth of the $2p$ state in hydrogen, so further improvements here will be very difficult.

In the last few years there has been an important improvement in measurements of the energy shift of the $1s$ ground state itself, by direct comparison of the frequency of the $1s$ – $2s$

resonance with four times the frequencies of the $2s-4s$ and $2s-4d$ two-photon resonances. These s and d states are much narrower than the $2p$ state, so these frequency differences can be measured more accurately than the classic Lamb shift. For a brief while it seemed that there was a discrepancy here between theory and experiment. Calculations [12,13] showed that for a proton radius $r_p = 0.862(11) \cdot 10^{-13}$ cm or $0.805(11) \cdot 10^{-13}$ cm, the inclusion of proton size and other corrections increases the theoretical $1s$ energy shift from the above result of 8127.4 MHz to 8173.12(6) MHz or 8172.94(9) MHz, respectively. This result for a proton radius $r_p = 0.862(11) \cdot 10^{-13}$ cm, which is believed to be more reliable, was not quite in agreement with the measured value [13] of 8172.86(5) MHz. But a later calculation [14] that adopts this proton radius and includes terms of order $\alpha^2(Z\alpha)^5$ yields energy shifts for the $1s$, $2s$, and $4s$ states in agreement with experiment. So apparently quantum electrodynamics wins again.

Problems

1. Consider a charged scalar particle of mass $m \neq 0$, described by a field $\phi(x)$ whose only interaction is with an external time-independent electromagnetic field $\mathcal{A}^\mu(x)$. Let $|N\rangle$ be a complete set of normalized one-boson or one-antiboson states with energies E_N , and define

$$u_N(\mathbf{x})e^{-iE_N t} = \langle \Omega | \phi(\mathbf{x}, t) | N \rangle, \quad v_N(\mathbf{x})e^{iE_N t} = \langle N | \phi(\mathbf{x}, t) | \Omega \rangle,$$

where $|\Omega\rangle$ is the vacuum. Show that the u_N and v_N together form a complete set, and give formulas for the coefficients of u_N and v_N in the expansion of a general function $f(\mathbf{x})$.

2. Suppose we include radiative corrections in the theory of Problem 1. Let $i\Pi^*(x, y)$ be the sum of all diagrams with one incoming and one outgoing charged scalar line (excluding final scalar propagators) to order α . Derive a formula for the shift in the energies E_N of the one-boson states due to these radiative corrections, in terms of $u_N(\mathbf{x})$ and $\Pi^*(x, y)$.
3. Use the results of Section 14.3 to calculate the radiative decay rate of the $2p$ states of hydrogen.
4. Suppose that the electron has an interaction with a light scalar field ϕ , of the form $g\phi\bar{\psi}_e\psi_e$. Suppose that the scalar mass m_ϕ is in the range $(Z\alpha)^2 m_e \ll m_\phi \ll Z\alpha m_e$. Calculate the change in the energy of the $1s$ state of hydrogenic atoms due to this interaction.
5. Carry out the calculation of Problem 4 for $m_\phi = 0$.

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